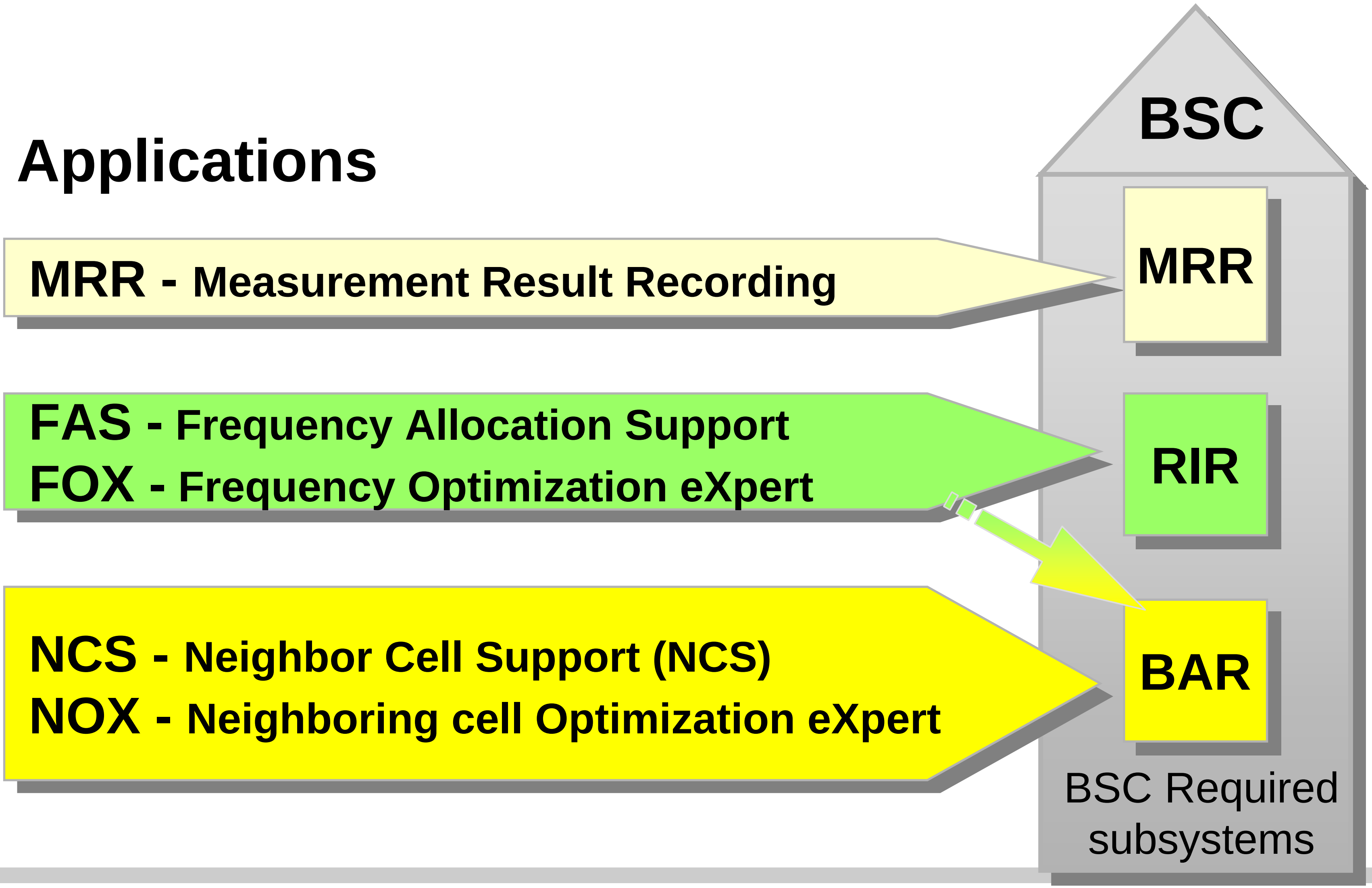
Agenda

- NCS/NOX
- Exercises

RNO - Radio Network Optimization



NCS/NOX

*** optimize correct neighbouing cell to be measure by MS to assist BSC in HO decision**

BA-List

- BCCH Allocation list.
- A list of BCCH frequencies for neighbouring cells.
- The idle BA-List contains all BCCH frequencies of an operator and is broadcast on the BCCH.
- The active BA-List is defined per cell and contains only BCCH frequencies for neighbouring cells.
- The active BA-List can be measured and improved with NCS.

BA-List Change Interval

- The time, in minutes, that a BCCH-frequency should be in the active BA-List each time.
- This is if more frequencies than the number of frequencies to be added, are included in the recording, each frequency will be in the active BA-List for the time stated, and then removed, to let the rest of the frequencies be included.
- When all frequencies have been in the BA-List, the first ones are included again. The value range is 10 to 1440 minutes (=1440 minutes is the same as 24 hours).

- Handover decisions are based on measurements on neighbouring cells, performed by the mobile station (MS) in active mode.
- A list of which frequencies to measure on is repeatedly sent downlink to the mobile throughout a call.
- This frequency list, the active mode BCCH Allocation (BA) list, is sent on the Slow Associated Control Channel (SACCH).
- The mobile receives it, measures signal strength and tries to decode the unique Base Station Identity code (BSIC) for all the frequencies defined in the list.
- As a result, the six strongest frequencies for which the BSIC was decoded successfully are reported back to the system in a measurement report, every 480 ms.
- The measurement report is sent on the SACCH (uplink).

- It is important to have the correct neighbouring cell relations in a network.
- A handover can only be made from one cell to another if the cells have a defined neighbouring cell relation.
- From this aspect, it is beneficial to have many defined neighbours.
- On the other hand, if too many cells are defined as neighbours to a cell, the active mode BA list may become too long.
- This will lead to reduced accuracy in the measurement reports from MSs.
- Both ways, quality will deteriorate and calls may be lost because a needed handover could not be made at the right time or to the correct cell.

- It is often difficult to find correct neighbouring cell relations.
- This is especially true in complex environments and in dense networks where it can be hard to predict what neighbours a cell should have.
- **From handover statistics one can find out if any neighbouring cell relations can be removed, but not if any specific relation should be added.**
- One way to get such information would be to add BCCH frequencies of potential neighbours to the active mode BA list of a cell, and observe how frequently the potential neighbours occur in measurement reports.
- The more frequently a potential neighbour is reported, the greater the reason to have it as a defined neighbour.

NSC/NOX uses:

- **Active BA list recordings (BARs) from BSC**
- **Handover statistics from OSS/SDM (**the collection of handover statistics in the system has to be set up outside the NCS/NOX application)
- **Network configuration data from OSS/CNA**
- **Control parameters form RNO/NCS/NOX**
- **Inputs specified by the user**

- The removal of neighbours is based on handover statistics collected from SDM.

- **If there has been very few handover attempts between two cells during a period stated by a system parameter (default is 15 days), and during the same time there has been enough outgoing handovers from the cell, so the data is reliable, then the relation should be removed.**

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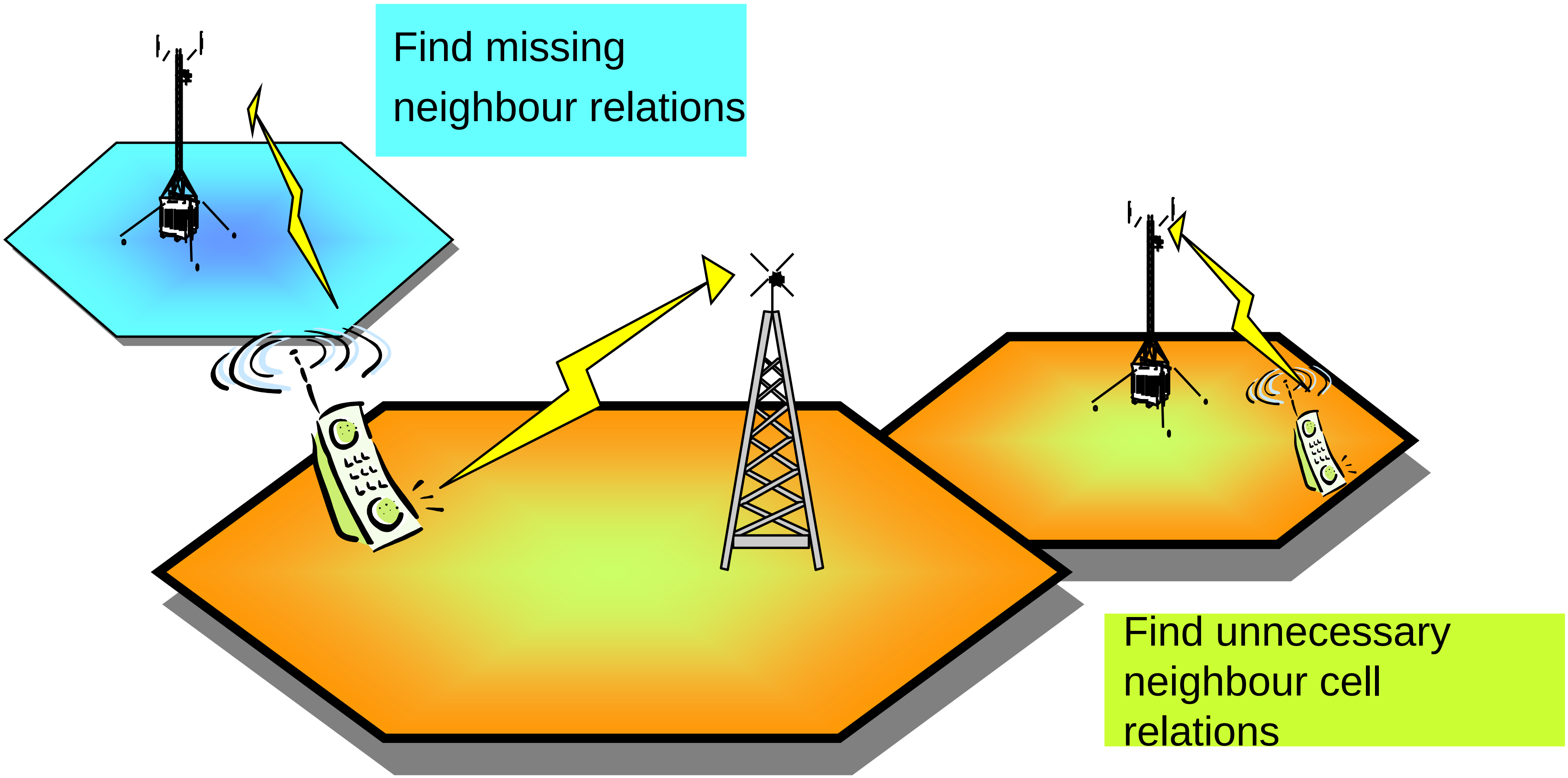
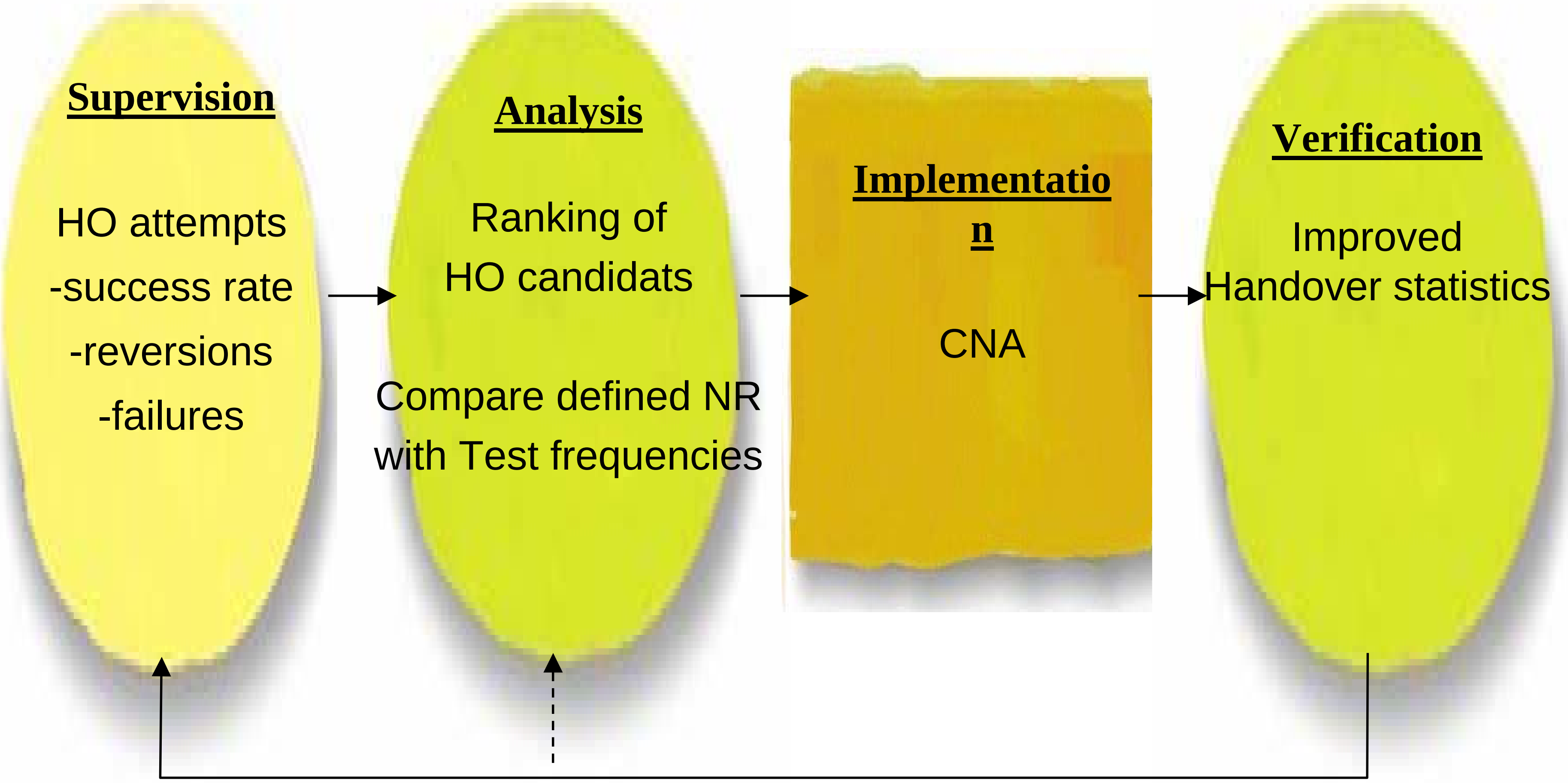
NCS/NOX - Neighbouring Cell Support/eXpert



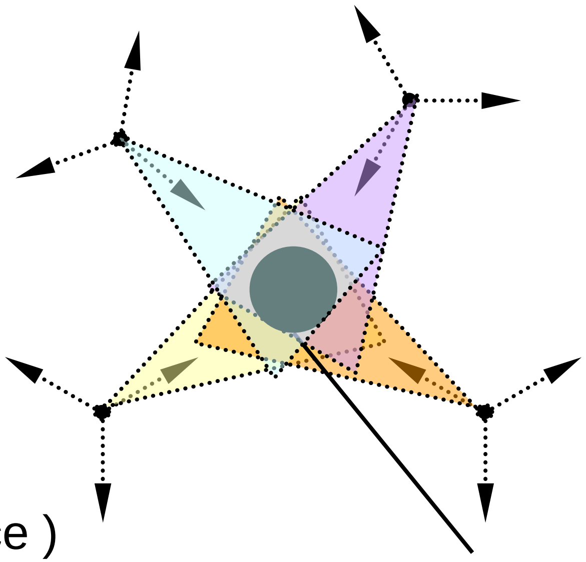
NCS/NOX - Optimization



NCS/NOX - Examples of usage

- Find new Neighbour Relations
- Remove unnecessary Neighbour Relations
- Areas with no dominant server
- Adding a new cell
- Cell coverage other than expected
- Troubleshooting in a cell

(too many dropped calls and bad HO performance)



“Circle of death”

NCS - Important Recording Parameters

Mutual Relation

- A neighbouring cell relation from a Cell A to a Cell B, and from a Cell B to a Cell A.

Single Relation

- A neighbouring cell relation that exists in only one direction, that is from a Cell A to a Cell B (but not from a Cell B to a Cell A).

Absolute Signal Strength Threshold

- **The lowest acceptable signal strength from the transmitter (BTS) that is reported from the mobile stations in the cells in the selected cell set**. The range is -110.5 to -47.5 dBm or 0 to 63 GSM.

- It is set to define the signal strength above which measurement reports are considered and evaluated compared to the signal strength of the serving cell. It is defined by the user per recording and can be thought of as corresponding to the handover hysteresis.

Relative Signal Strength Threshold

- The difference in signal strength between the signal strength of a neighbouring cell and the serving cell. The value range is 0 - 63 dB.

- **SS rel=SSneighb-SSserv**

- If the signal strength for the neighbouring cell is higher than or equal to this threshold value, the occurrence of the neighbouring cell is recorded (counted) as an absolute signal strength occurrence. It is however not used in the decision making regarding cell relations in NCS/NOX.

The two criteria's NCS used to find new Neighbours

- **The relative signal strength threshold** is related to a counter that is incremented every time equation (1) is fulfilled: **SS f,BSIC > SSserv + rel (1)**
- In equation (1) SSf,BSIC is the reported signal strength for that frequency and BSIC, SSserv is the reported signal strength for serving cell and rel is the relative signal strength threshold. For each cell, there is one counter for every reported frequency/BSIC combination. The threshold can preferably be set to the handover hysteresis in Locating.

- **The absolute signal strength threshold** is related to a counter that during the recording is incremented every time equation (2) is fulfilled; **SS f,BSIC > abs (2)** where abs is the absolute signal strength threshold.
- For each cell, there is one counter for every reported frequency/BSIC combination. The threshold can preferably be set to the threshold used between layers in the Hierarchical Cell Structures feature (see ref. 2), and the counter then indicates whether many handovers would have been made. This only applies to cells in different layers of the hierarchical cell structure, from a cell in a higher layer to a cell in a lower layer.

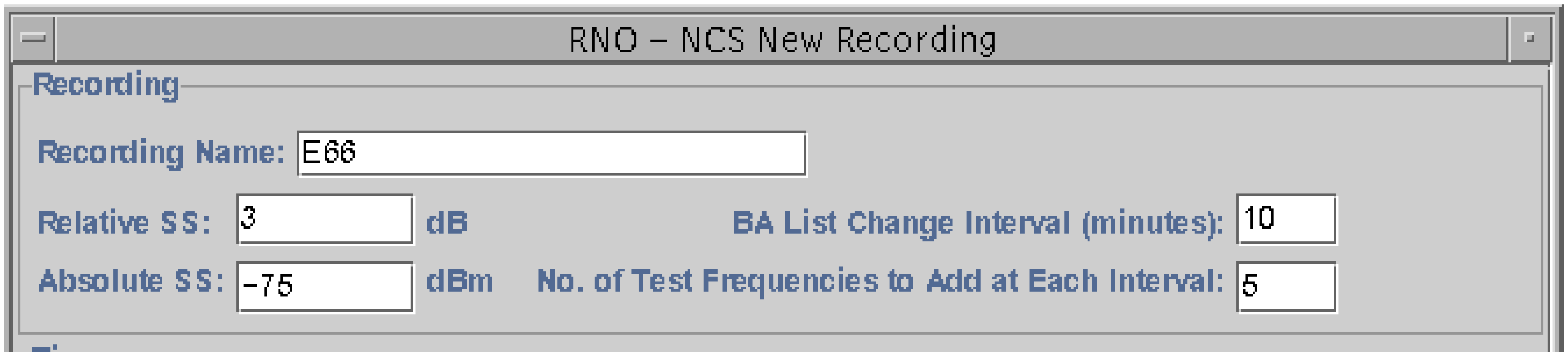
1st Order Neighbour

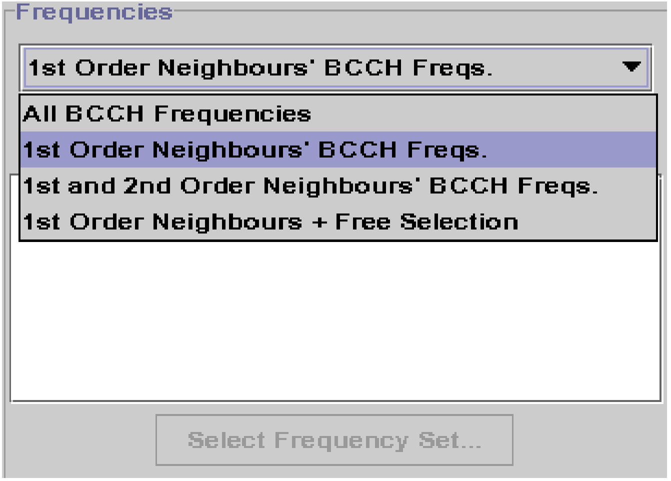
- The BCCH frequencies that are currently defined as neighbours.

2nd Order Neighbour

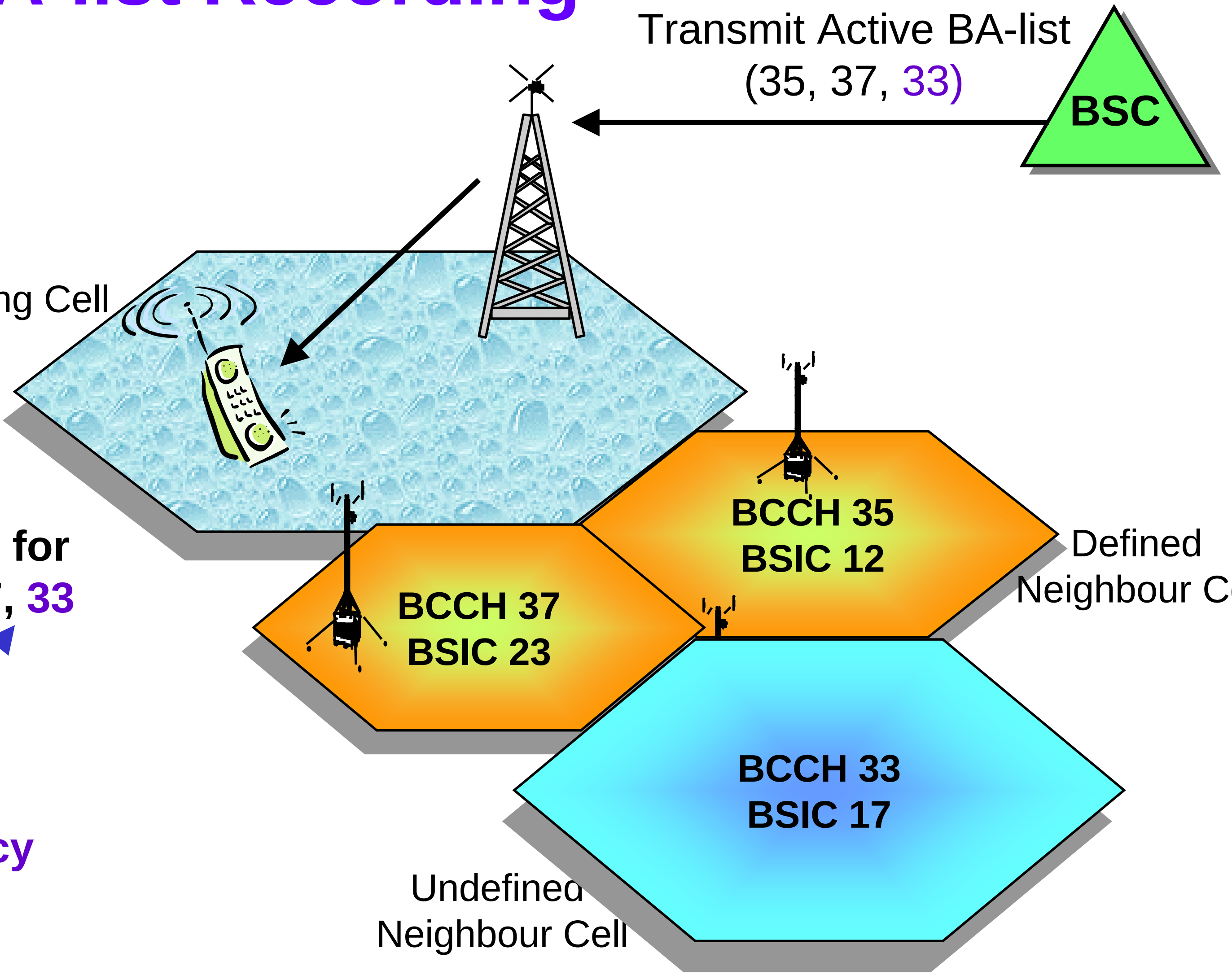
- The neighbouring cells to the neighbours of the defined cell.

NCS - Important Recording Parameters





Active BA-list Recording



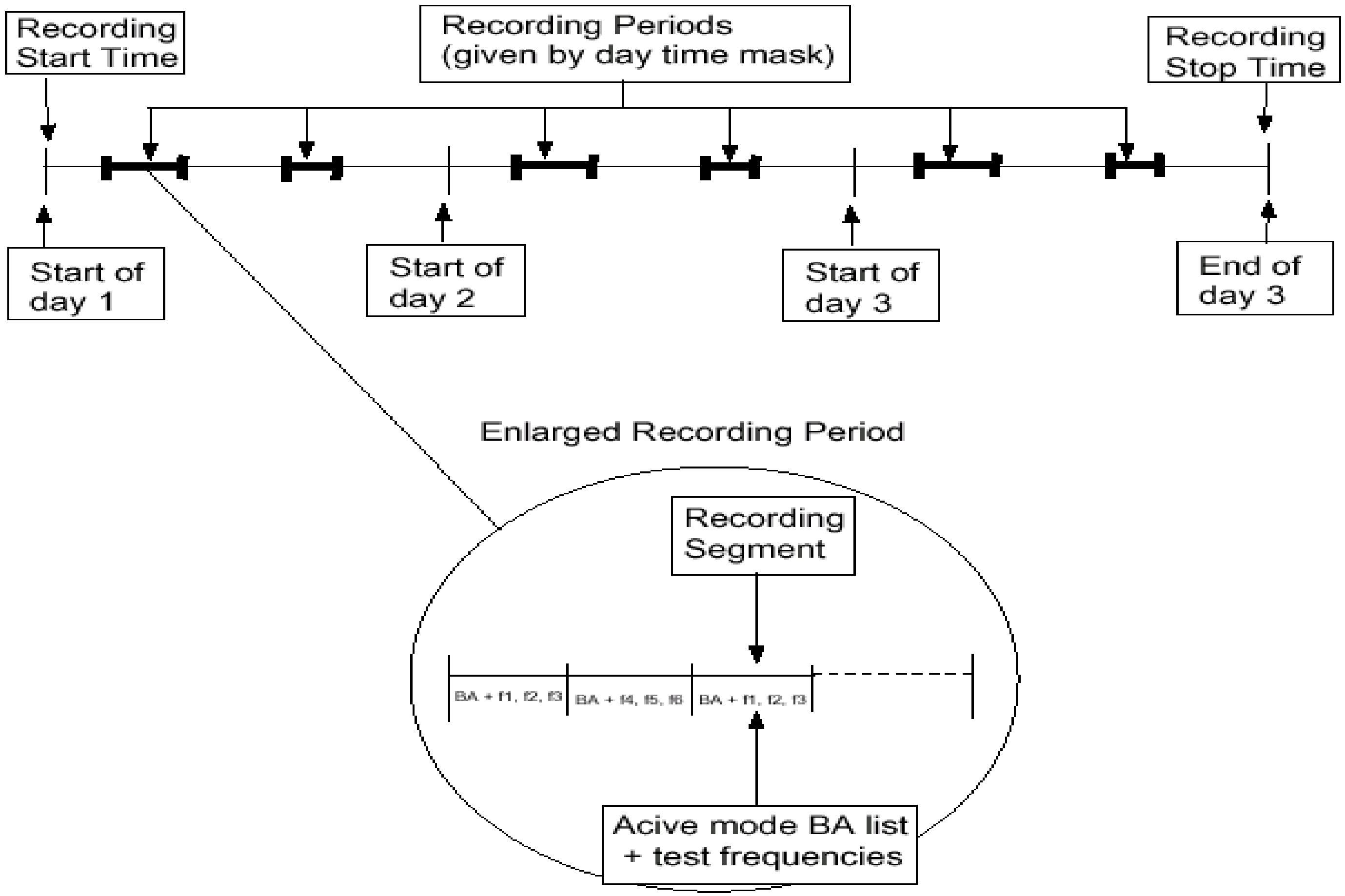
Mobile listen for BCCH 35, 37, 33

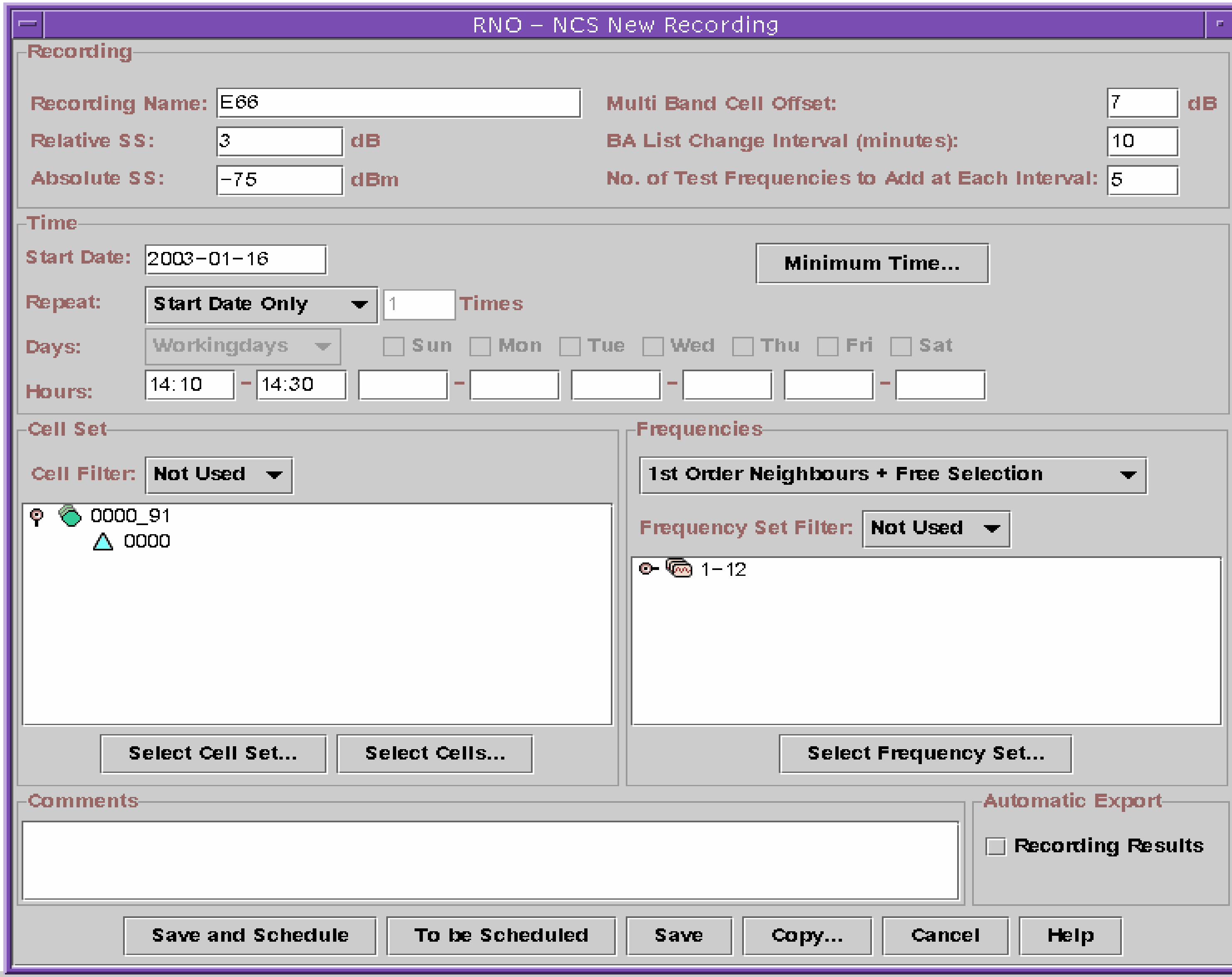
Test frequency

BAR

- OSS/NCS user can temporarily add other frequencies (test frequencies) to the Active BA-List and get measurement results from these test frequencies as well.
- The BSC Recording function active BA list Recording (BAR) adds BCCH-carriers, to the active BA list in a group of cells selected by the user.
- BAR then records, among other things, the average signal strengths on each of these frequencies with BSICs.
- The data is taken from Measurement Reports sent by active mobiles in the measuring cells.

Recording Session



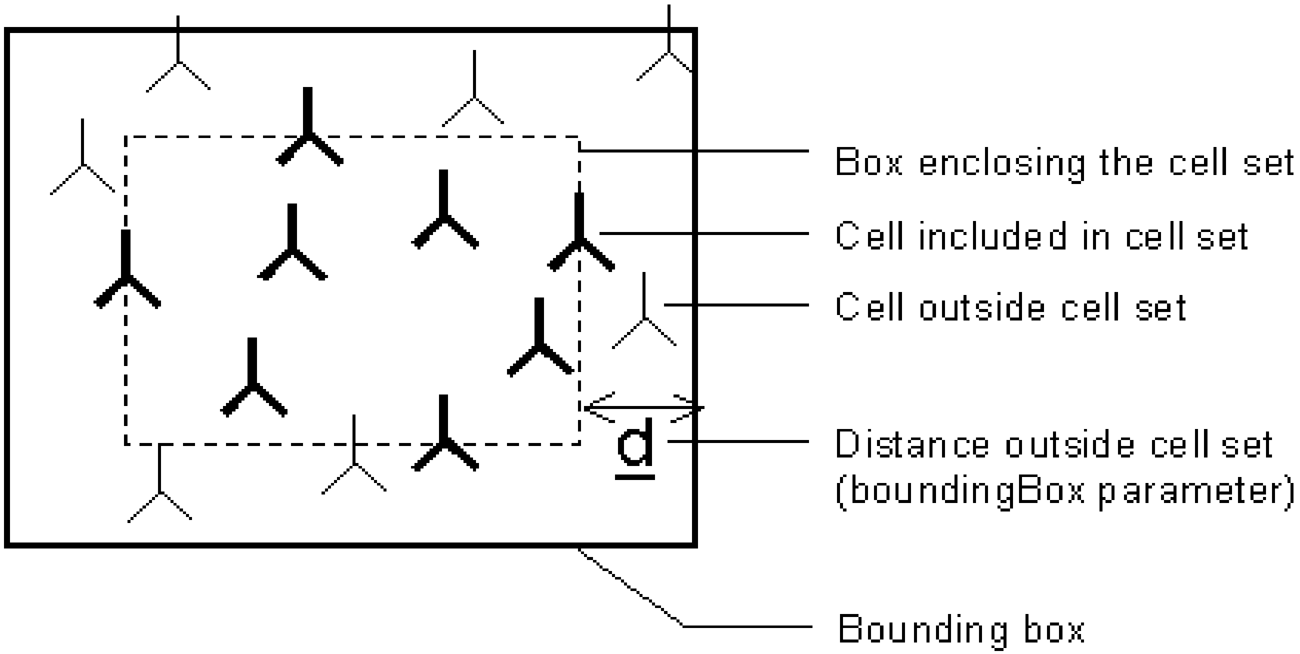

RNO – NCS New Recording
Recording
Recording Name: E66
Relative SS: 3 dB
Absolute SS: -75 dBm
Multi Band Cell Offset: 7 dB
BA List Change Interval (minutes): 10
No. of Test Frequencies to Add at Each Interval: 5
Time
Start Date: 2003-01-16
Minimum Time...
Repeat: Start Date Only
1 Times
Days: Workingdays
Sun Mon Tue Wed Thu Fri Sat
Hours: 14:10 - 14:30
Cell Set
Cell Filter: Not Used
0000_91
0000
Select Cell Set...
Select Cells...
Frequencies
1st Order Neighbours + Free Selection
Frequency Set Filter: Not Used
1-12
Select Frequency Set...
Comments
Automatic Export
Recording Results
Save and Schedule
To be Scheduled
Save
Copy...
Cancel
Help

The bounding box

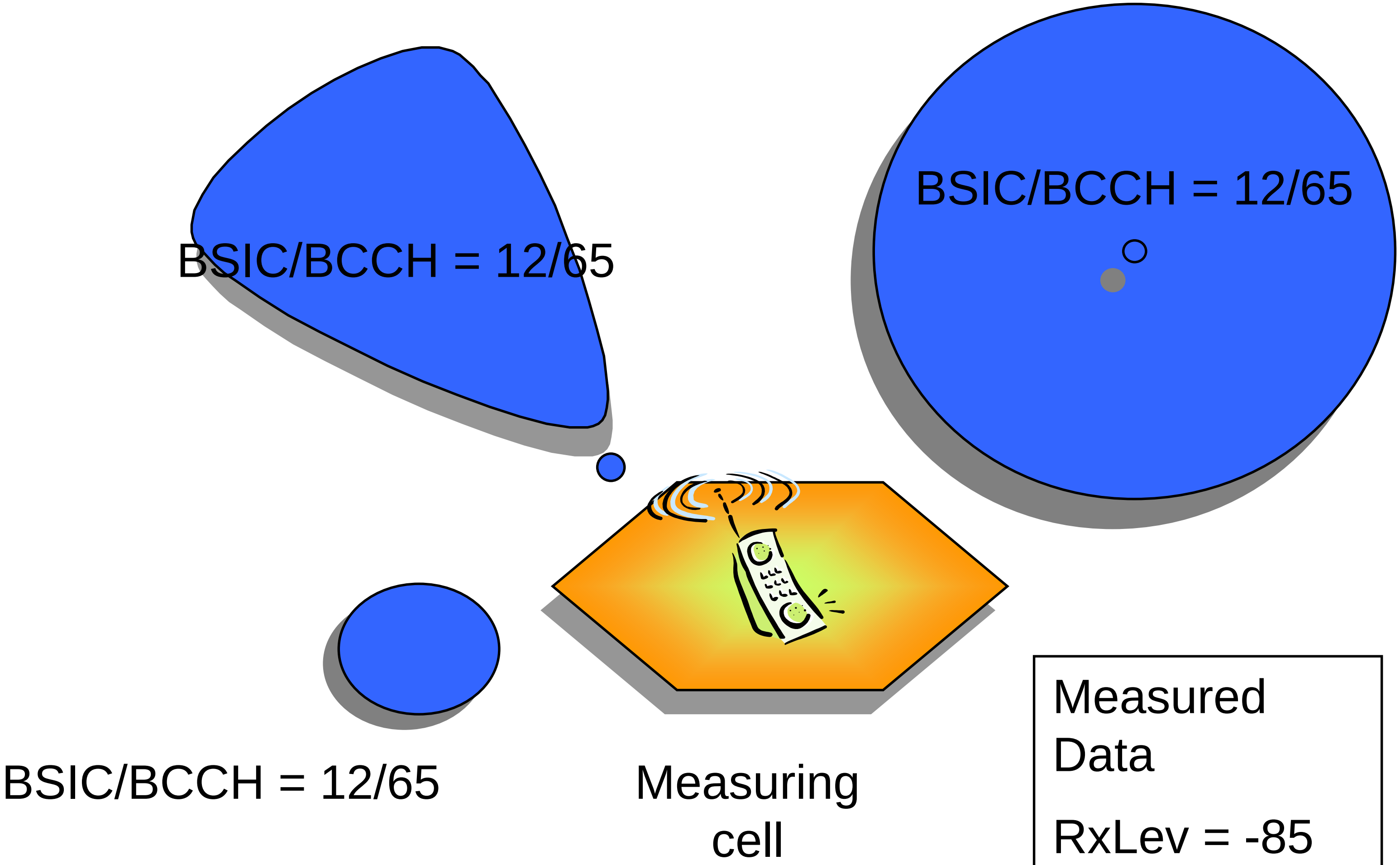
- The area, in which the application searches for the probable cell name, is the area of the set union of:
  - Cells in the cell set
  - First order neighbours
  - Second order neighbours
  - Cells within the Bounding Box

The bounding box

- is a thought rectangle that encloses all cells in the cell set plus the cells within the area from a distance ***d*** that can be 0 - 100 kilometers outside the cell set as shown in the following figure:



NCS/NOX - Finding probable cell



Measured Data

RxLev = -85
BSIC = 12
BCCH = 65

NCS - Reports

- Overview Report
- Cell Report
- Detailed Cell Report
- Cell Report Chart
- Change Order
- Probable Neigbouring Cell Report

ERICSSON

NCS - Cell Report

RNO – NCS Cell Report

File View Reports Tools Help

Recording Name: nox 0418 1800:1
Cell Name: VABSC1/VAL1113
No. of BA List Frequencies: 11
HCS Layer:

Relative SS Threshold: 3 dB
Absolute SS Threshold: -75.0 dBm
Creator: ericsson
HCS Layer Threshold:

Defined Neighbours

Cell Name	BCCH Freq.	BSIC	BA List Time (min)	No. of Measurement Reports	Reports Above Rel. SS Threshold (%)	Reports Ranked as No. 1 (%)	Average SS when No. 1 (dBm)	No. of HO Attempts	Success (%)	Reversions (%)	Failures (%)
VABSC1/VAL1102	516	53	197	27,118	0.03	0.00	-91.50	2,971	98.25	1.21	0.54
VABSC1/VI1191	90	62	197	27,118	0.07	0.11	-81.50	144	91.67	8.33	-0.00
VABSC1/VAL1101	514	50	197	27,118	0.08	0.06	-88.50	294	90.14	9.18	0.68
VABSC1/VA1052	72	62	197	27,118	0.53	0.85	-77.50	397	97.23	2.27	0.50
VABSC1/VAL1032	520	51	197	27,118	0.60	1.32	-91.50	2,437	94.09	5.38	0.53
VABSC1/VI1062	78	54	197	27,118	1.80	2.08	-80.50	2,961	98.14	1.18	0.68
VABSC1/VAL1112	538	50	197	27,118	2.77	3.58	-78.50	1,289	96.59	2.64	0.78
VABSC1/VAL1111	524	53	197	27,118	7.00	9.40	-80.50	4,930	94.79	4.75	0.47
VABSC1/VI1203	74	53	197	27,118	11.45	49.61	-76.50	20,984	98.82	0.59	0.59

Undefined Neighbours

Probable Cell Name	BCCH Freq.	BSIC	BA List Time (min)	No. of Measurement Reports	Reports Above Rel. SS Threshold (%)	Reports Ranked as No. 1 (%)	Average SS when No. 1 (dBm)
VABSC1/VI1083	70	65	57	7,377	9.94	22.81	-78.50
VABSC1/VI1201	92	64	50	7,176	8.05	18.60	-78.50
VABSC1/VA1012	90	54	197	27,118	3.83	7.67	-80.50
VABSC1/VA1410	50	64	57	7,679	0.72	0.36	-90.50
VABSC1/VAL1092	534	51	197	27,118	0.11	0.11	-92.50
VABSC1/VA1082	84	66	197	27,118	0.10	0.04	-88.50
VABSC1/VI1061	88	66	50	6,829	0.10	0.01	-80.50
VABSC1/VA1022	72	74	197	27,118	0.10	0.12	-78.50
VABSC1/VI1082	78	55	197	27,118	0.07	0.32	-89.50

Exporting Data to CNA

- To update the network, the changes stated in an NCS Change Order dialogue box are exported to CNA, either via a CNAI-file or to a Planned Area so that a CNA operator can execute the changes.

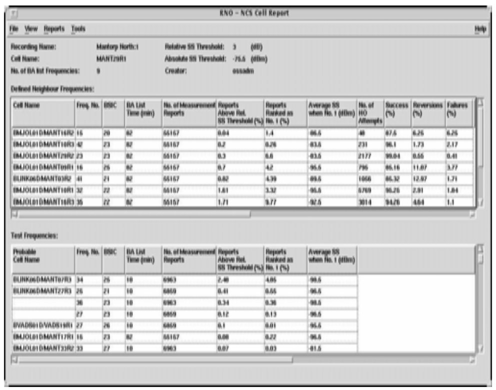
- Exporting the change to a CNAI-file brings up a standard file selection dialogue box.

- You select undefined neighbours to be added to the BA-List and defined neighbouring relation to be removed by selecting them in the NCS Cell Report list and add them to a NCS Change Order by clicking on File -> Add (Additions/Removals) to Change Order.

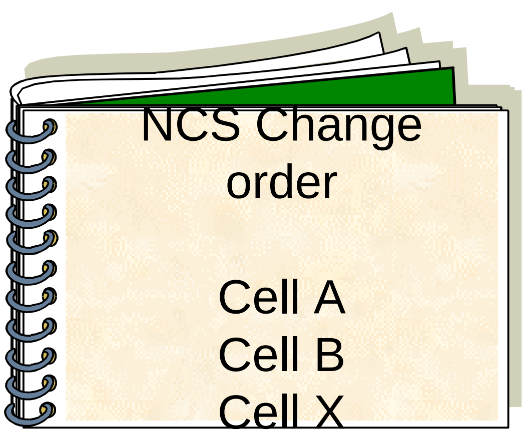
- The CNA operator takes care of the CNAI-file or the Planned Area when he or she plans to update the network.

NCS - Change Order (new BA list to CNA)

NCS Cell Report

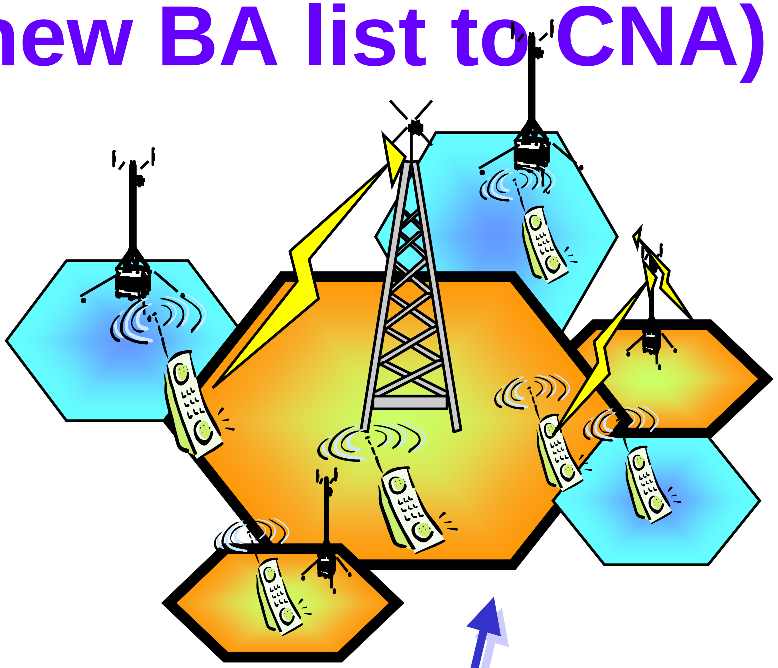


Add Neighbour to Change order



Export to CNA planned area

Update network



Export new neighbour to CNA

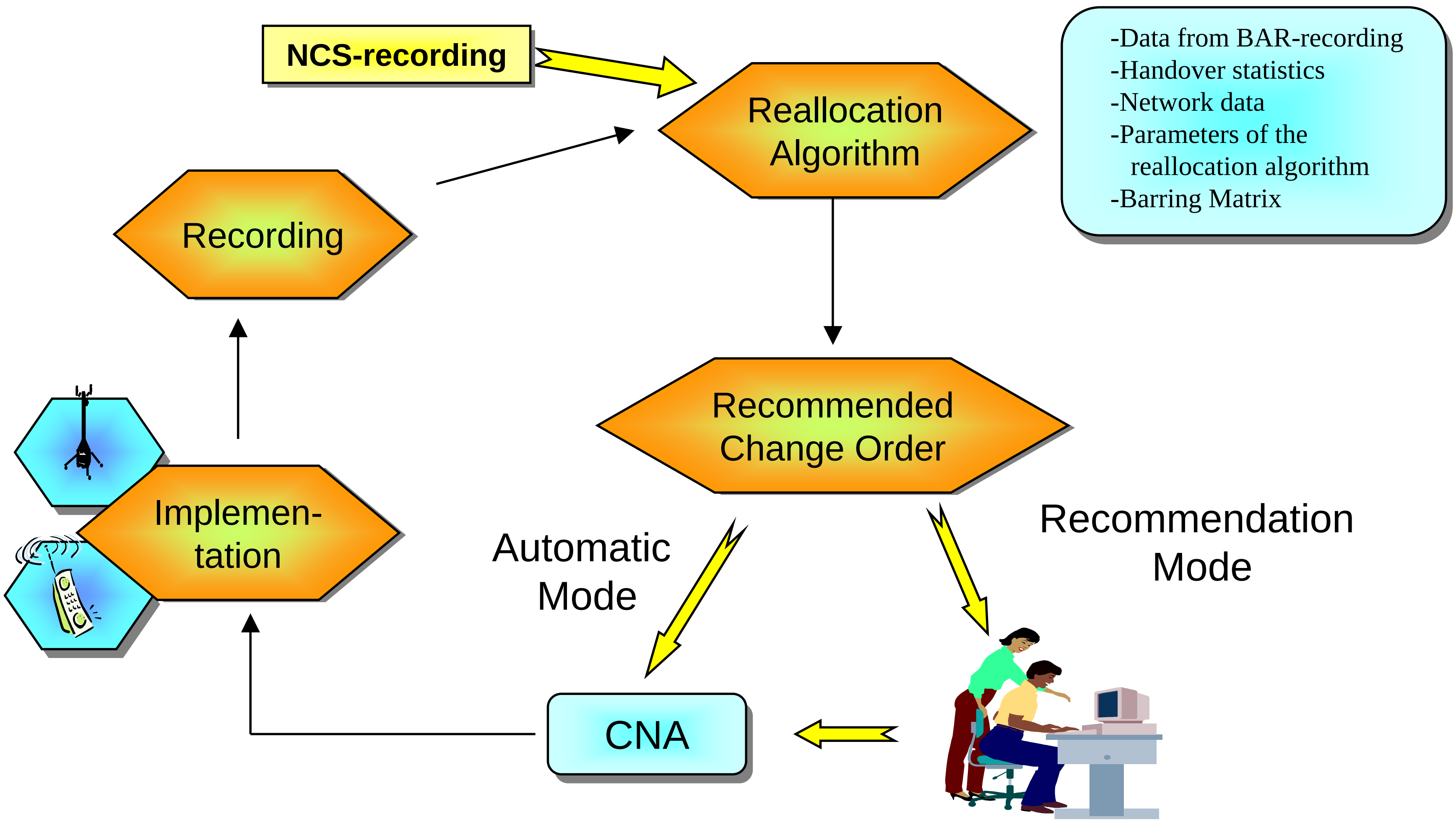
RNO – NCS Change Order

Neighbouring Cells to Add | Neighbouring Cells to Remove

Measuring Cell	BCCH Freq.	BSIC	Neighbouring Cell to Add	No. of Measurement Reports	Reports Above Rel. SS Threshold (%)	Reports Ranked as No. 1 (%)	Average SS when No. 1 (dBm)
AV00/AV00071	49	30	AV00/AV00055	40000	6.90	15.10	-65.50
AV00/AV00071	518	31	AV00/AV00082	20000	24.14	39.98	-74.50

Export to CNAI File... | Export to Planned Area... | Delete | Delete All | Close | Help

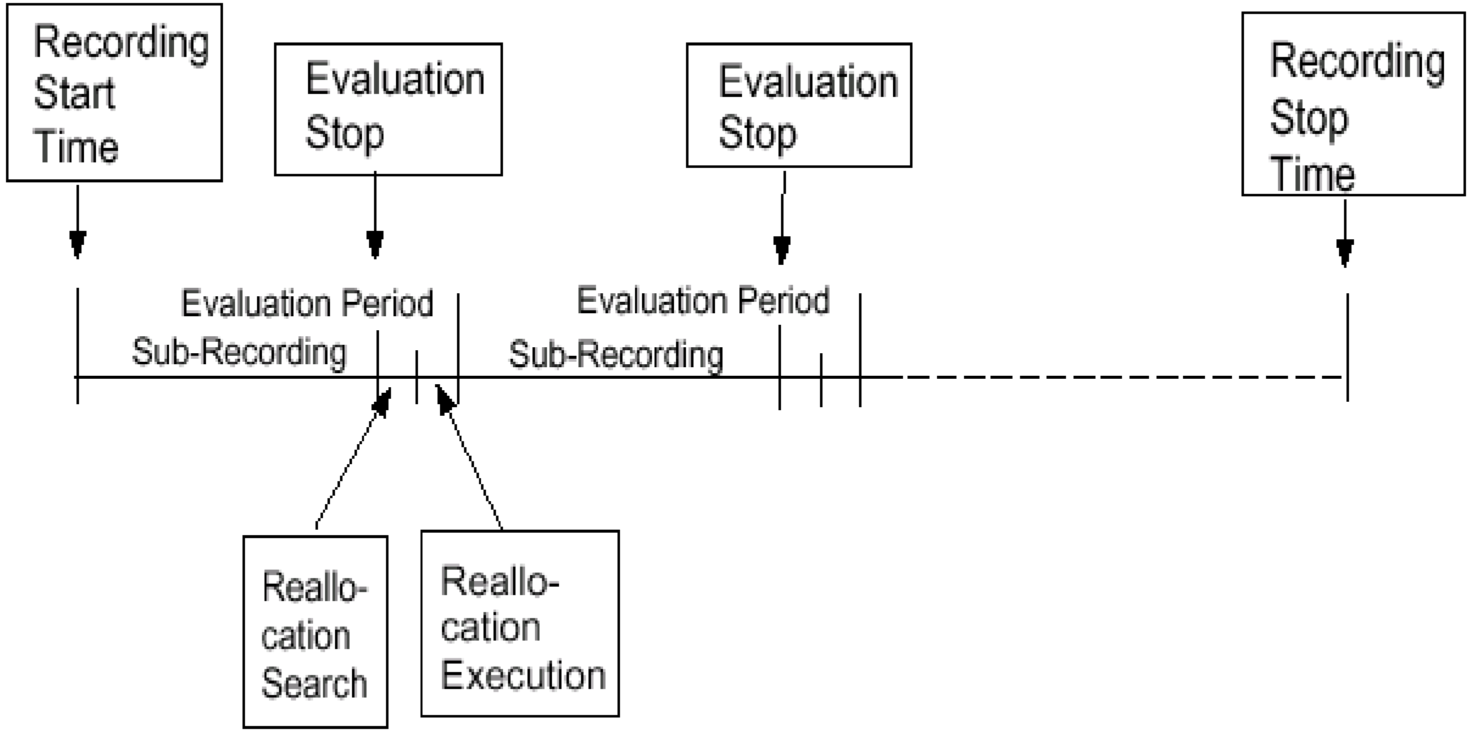
NOX - Neigbouring cell list Optimization eXpert



Barring Matrix

- The Barring Matrix is not handled via the graphical user interface of NOX.
- The Barring Matrix consists of two ASCII-text files located in the /var/opt/ericsson/ncs/data/db/ directory, and is **used to specify which neighbouring cell relations that should be mandatory or barred.**
- The name of the files are barredRelations.txt and mandatoryRelations.txt.
- The barredRelations.txt file contains relations that are barred for addition and the mandatoryRelations.txt file contains relations that are barred for removals.

Recording Session - Recommended or automated mode



NOX - Reports

• Change Order Recommendation

It is possible to modify the suggestions in the NOX Change Order Recommendation.

The Change Order Recommendation can then be exported to a CNAI-file or to a Planned Area.

• Change Order Evaluation

•get information about approved reallocations

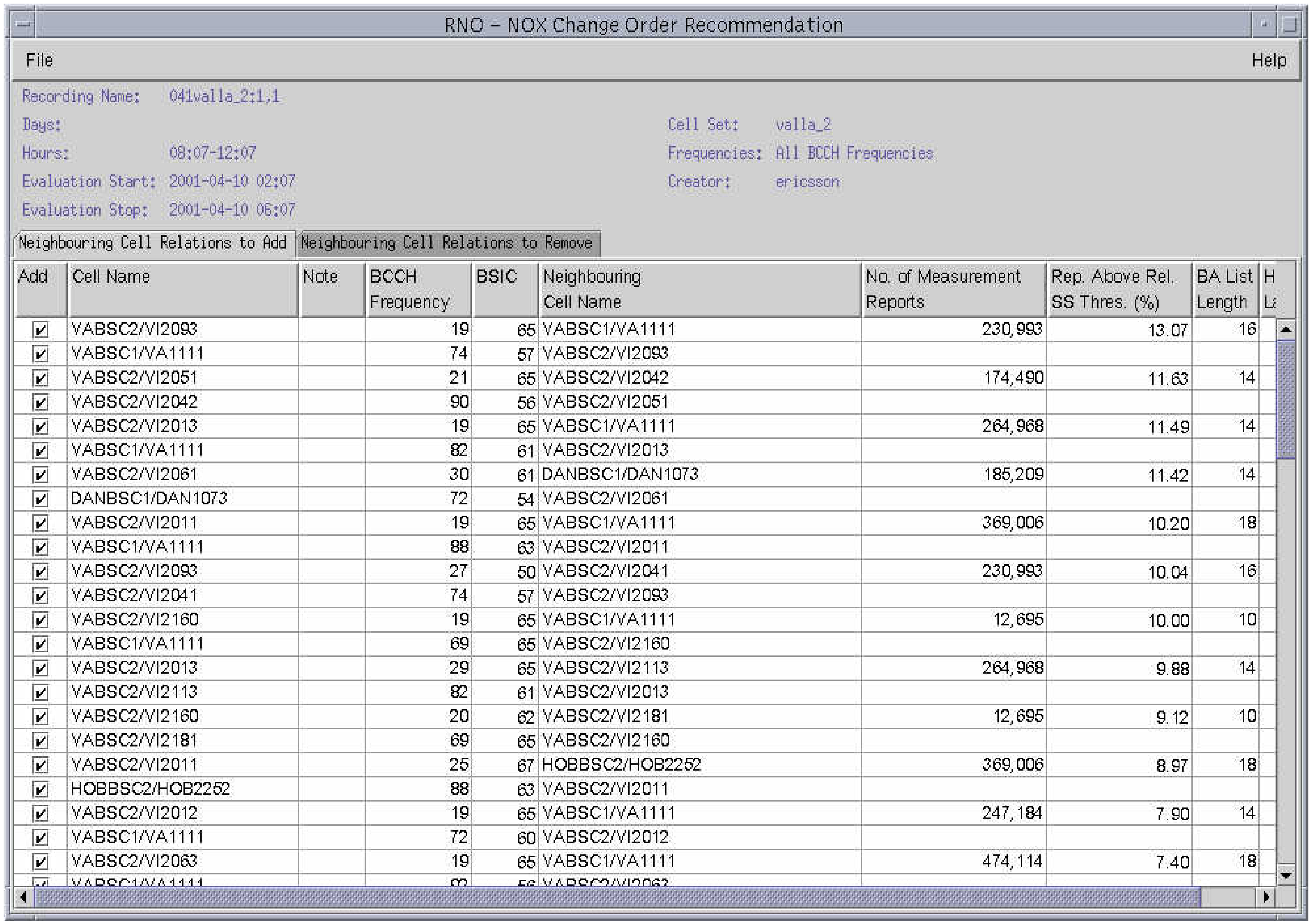
The Change Order Evaluation is generated when the user accepts the recommended

changes or when an NOX Automatic recording has been executed.

•(Reallocation Log)

contains information about how many neighbouring cell relations that were ordered to be added and/or removed from the network, and how many of them that were really implemented and when these changes took place

NOX - Change Order Recommendation



RNO – NOX Change Order Recommendation

File | Help

Recording Name: 041valla_2;1,1
Days:
Hours: 08:07-12:07
Evaluation Start: 2001-04-10 02:07
Evaluation Stop: 2001-04-10 06:07

Cell Set: valla_2
Frequencies: All BCCH Frequencies
Creator: ericsson

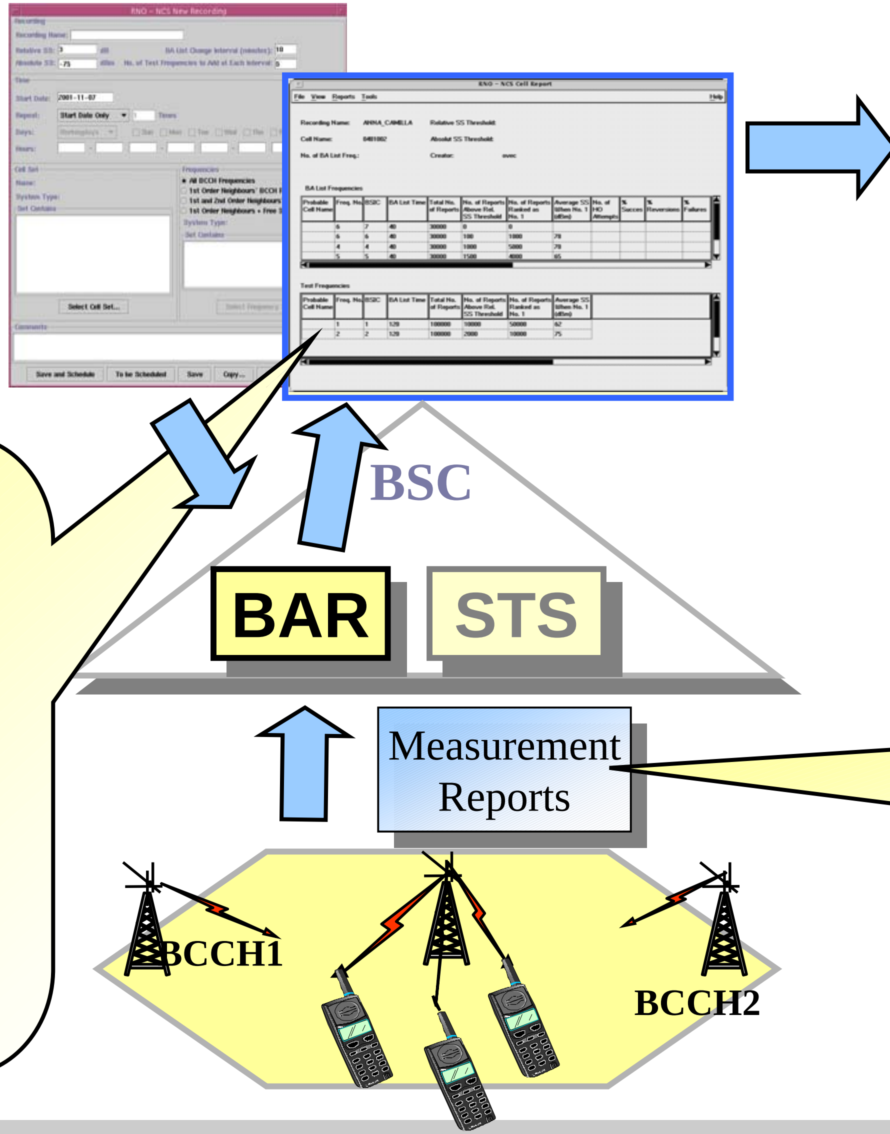
Neighbouring Cell Relations to Add | Neighbouring Cell Relations to Remove

Add	Cell Name	Note	BCCH Frequency	BSIC	Neighbouring Cell Name	No. of Measurement Reports	Rep. Above Rel. SS Thres. (%)	BA List Length	H L
☑	VABSC2/VI2093		19	65	VABSC1/VA1111	230,993	13.07	16	
☑	VABSC1/VA1111		74	57	VABSC2/VI2093				
☑	VABSC2/VI2051		21	65	VABSC2/VI2042	174,490	11.63	14	
☑	VABSC2/VI2042		90	56	VABSC2/VI2051				
☑	VABSC2/VI2013		19	65	VABSC1/VA1111	264,968	11.49	14	
☑	VABSC1/VA1111		82	61	VABSC2/VI2013				
☑	VABSC2/VI2061		30	61	DANBSC1/DAN1073	185,209	11.42	14	
☑	DANBSC1/DAN1073		72	54	VABSC2/VI2061				
☑	VABSC2/VI2011		19	65	VABSC1/VA1111	369,006	10.20	18	
☑	VABSC1/VA1111		88	63	VABSC2/VI2011				
☑	VABSC2/VI2093		27	50	VABSC2/VI2041	230,993	10.04	16	
☑	VABSC2/VI2041		74	57	VABSC2/VI2093				
☑	VABSC2/VI2160		19	65	VABSC1/VA1111	12,695	10.00	10	
☑	VABSC1/VA1111		69	65	VABSC2/VI2160				
☑	VABSC2/VI2013		29	65	VABSC2/VI2113	264,968	9.88	14	
☑	VABSC2/VI2113		82	61	VABSC2/VI2013				
☑	VABSC2/VI2160		20	62	VABSC2/VI2181	12,695	9.12	10	
☑	VABSC2/VI2181		69	65	VABSC2/VI2160				
☑	VABSC2/VI2011		25	67	HOBBSC2/HOB2252	369,006	8.97	18	
☑	HOBBSC2/HOB2252		88	63	VABSC2/VI2011				
☑	VABSC2/VI2012		19	65	VABSC1/VA1111	247,184	7.90	14	
☑	VABSC1/VA1111		72	60	VABSC2/VI2012				
☑	VABSC2/VI2063		19	65	VABSC1/VA1111	474,114	7.40	18	

NCS Recording Capacity

Resource	Capacity	Description
Number of simultaneous recordings under one OSS	64 /BSC	-
Number of simultaneous recordings in one BSC	64	-
Number of cells in one Cell Set	2000	-
Number of frequencies in one recording	No limit	It is possible to include all ARFCNs in the system
Number of Cell Sets that can be defined	100	-
Number of Frequency Sets that can be defined	100	-
Number of stored results	100	-
Duration of one recording	31 days	The time schedule in one recording configuration may be 31 days long. The recording in the BSC can not be longer than one day, which means that new BSC recordings are started each day.

NCS - Addition of neighboring cell relation - Summary



Addition of neighboring cell relation

Not defined neighbor, but a good candidate.

Add to BA-list!

Measure signal strength of BCCH in BA-list of:

- Defined neighbors
- Test frequencies

Statistics on BCCH level:

- Number of reports above the relative signal strength threshold.
- Number of reports ranked as no.1
- Average signal strength when no.1

Statistics on cell level:

- Number of reports the best test frequency was ranked no.1
- Number of reports the worst configured frequency was ranked no.1